# Identifying Fashion Product Categories with Potential in the Taiwan Market

A Data Analytics Project Inspired by Global Fashion Trends and Cross-Border Shopping

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|---|---|
| Tools | Python, Microsoft Excel (Solver), Tableau Dashboard |
| Primary Dataset | Fashion Retail Sales Dataset |
| Secondary Dataset Framework | Global Fashion Brands Dataset for Future Brand-Level Integration |

## Executive Summary

This project analyzes fashion retail transaction data to identify product categories that may have market potential in Taiwan.  The project is inspired by global fashion trends and cross-border shopping behavior. This project analyzes fashion retail transaction data to identify product categories that may have market potential in Taiwan. Since the primary dataset does not contain brand information, the findings should be interpreted as category-level insights rather than brand-level recommendations.

The analysis uses 3,400 raw purchase records. After removing missing values and abnormal purchase amount, 2,453 clean records are used for category analysis. The strongest category by purchase frequency is Bags & Accessories, followed by tops, Outwear, Bottoms, Dresses & One-Piece, Footwear, and Longwear. Taiwan's Market Potential Score is created using purchase frequency, average purchase amount, and average customer rating.

The results suggest that Bags & Accessories, Footwear, and Tops should be prioritized for cross-border fashion product selection. For a stronger brand level version, this report proposes integrating a Global Fashion Brands dataset containing brand name, country of origin, region, ranking, brand equity, and growth information.

## 1. Business background

Cross-border shopping and international e-commerce have made global fashion products more accessible to consumers. For consumers in Taiwan, fashion product selection can be influenced by brand image, product type, price range, and perceived lifestyle value. However, from a business analysis perspective, product selection should rely not only on personal assumptions. A data-driven approach can help identify which fashion product categories show stronger demand signals and which categories may be worth testing in a cross-border retail context.

This project was developed as a business analysis portfolio project for Business Analytics. Instead of claiming direct business operation experience, the project uses public datasets and analytics methods to study fashion demand patterns and support cross-border product selection decisions.

## 2. Business Problem

A common challenge in cross-border fashion retail is deciding which products are worth selecting for a target market. Without transaction data or consumer preference analysis, product selection can become subjective. Retailers may over-focus on products that seem popular globally but do not match the target market, price range, or category demand.

The business problem addressed in this project is: How can data analytics help identify high potential product categories and propose a future framework for brand-level analysis.

## 3. Project Objective

The objective of this project is to analyze fashion retail data, identify high-potential product categories, and build a business recommendation framework for cross-border fashion product selection in Taiwan.

## 4. Research Questions

- Which fashion product categories show the strongest purchase demand?
- Which categories have a higher average purchase amount and customer ratings?
- Which product categories may have potential in the Taiwan market?
- How can brand-level data be integrated into future versions of this project?

## 5. Data Description

Primary dataset: Fashion Retail Sales Dataset. The uploaded transaction file contains six columns: Customer Reference, Item Purchased, Purchase amount, Date purchase, Review Rating, and Payment Method. The Kaggle page describes this dataset as containing 3,400 records of fashion retail sales with details about customer purchases.

**Table 1. Dataset Profile**

| Metric | Value |
|---|---|
| **Raw records** | **3,400** |
| **Raw columns** | **6** |
| **Clean records used** | **2,453** |
| **Missing purchase amount** | **650** |
| **Missing review rating** | **324** |
| **Unique items** | **50** |
| **Date range** | **2022 / 10 / 2 to 2023 / 10 / 1** |

Secondary dataset framework: Fashion brands appearing in the Interbrand Top 100 Global Brands lists between 2001 and 2021 form the basis of the mentioned public data set, along with details like Brand name, Country of origin, Region, Sub-sector in industry, Brand Ranking, Brand equity, and Growth in brand equity. Since no data set of the brand CSV file was uploaded, this analysis does not compute any score at the brand level.

## 6. Methodology

The project follows a standard analytics workflow: data understanding, data cleaning, feature engineering, exploratory data analysis, scoring model design, and business recommendation.

**Raw Retail Dataset**

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**Data Cleaning**

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**Feature Engineering**

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**Exploratory Data Analysis**

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**Taiwan Market Potential Score**

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**Business Recommendations**

I. Convert the purchase data into a valid datetime field.
II. Create broader product category groups from item names.
III. Remove rows with missing purchase amounts or missing review ratings.
IV. Remove abnormal purchase-amount outliers using the IQR rule.
V. Aggregate purchase frequency, total sales, average purchase amount, and average review score by product category.
VI. Calculate Taiwan Market Potential Score using normalized frequency of purchase, average purchase amount, and average review score.

## 7. Data Cleaning

Missing values exist in Purchase Amount and Review Ratings variables. Extreme Purchase Amount outliers have been found through the IQR technique and removed to avoid overstating category performance.

**Table 2. Data Cleaning Summary**

| Cleaning Step | Action | Reason |
|---|---|---|
| Handle missing purchase amount | Dropped from clean analysis | Unable to calculate revenue metrics |
| Handle missing review ratings | Dropped from clean analysis | Unable to calculate satisfaction metrics |
| Convert data format | Converted to datetime | Needed for monthly trend analysis |
| Create product categories | Mapped 50 items into 7 broader categories | Needed for business-level interpretation |
| Remove outliers | Removed amount outliers above IQR threshold | Avoid distorted sales results |

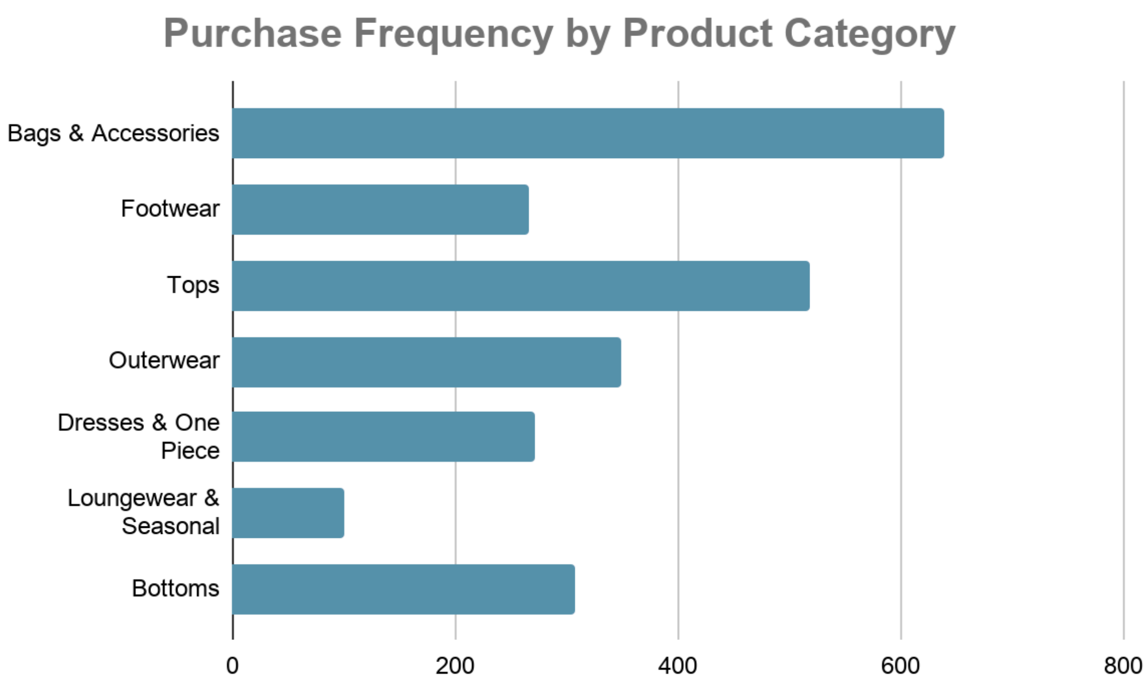
## 8. Exploratory Data Analysis

The product category analysis shows that Bags & Accessories and Tops have the highest purchase frequency. Bags & Accessories also has the strongest final potential score because it combines high purchase volume with a reasonable average purchase amount and customer ratings.

**Table 3. Category Summary and Potential Score**

| Product Category | Purchases | Total Sales (USD) | Average Purchase (USD) | Average Rating | Taiwan Potential Score |
|---|---|---|---|---|---|
| Bags & Accessories | 639 | 64,609 | 101.1 | 3 | 69 |
| Footwear | 267 | 29,470 | 110.4 | 3 | 60.4 |
| Tops | 519 | 57,012 | 109.8 | 2.9 | 50.5 |
| Outerwear | 349 | 37,543 | 107.6 | 3 | 50.3 |
| Dresses & One-Piece | 271 | 28,530 | 105.3 | 3 | 47.7 |
| Loungewear & Seasonal | 100 | 11,519 | 115.2 | 3 | 43 |
| Bottoms | 308 | 31,531 | 102.4 | 3 | 39.6 |

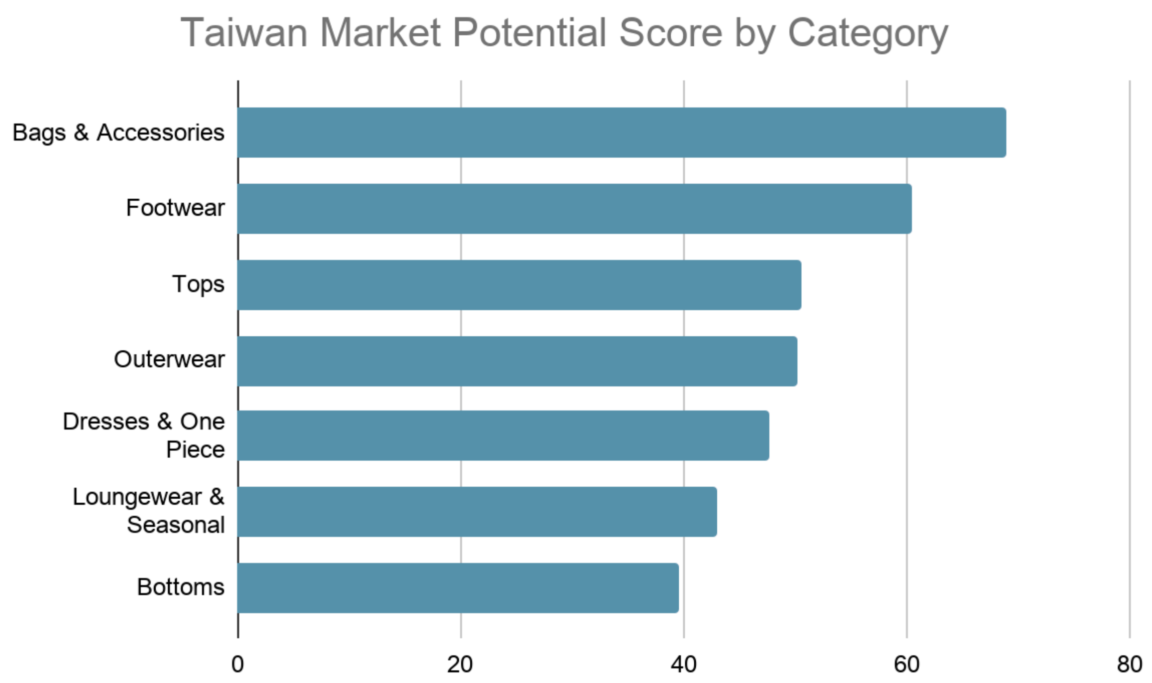
**Figure 1. Purchase Frequency by Product Category**




**Interpretation**

As can be seen in Figure 1 below, the purchase frequency was spread out in seven categories of products. Bags & Accessories witnessed the highest frequency of purchases, while Tops and Outerwear witnessed the second highest. However, Loungewear & Seasonal witnessed the lowest frequency of purchases. This means that accessories and general wear products were more popular among consumers.

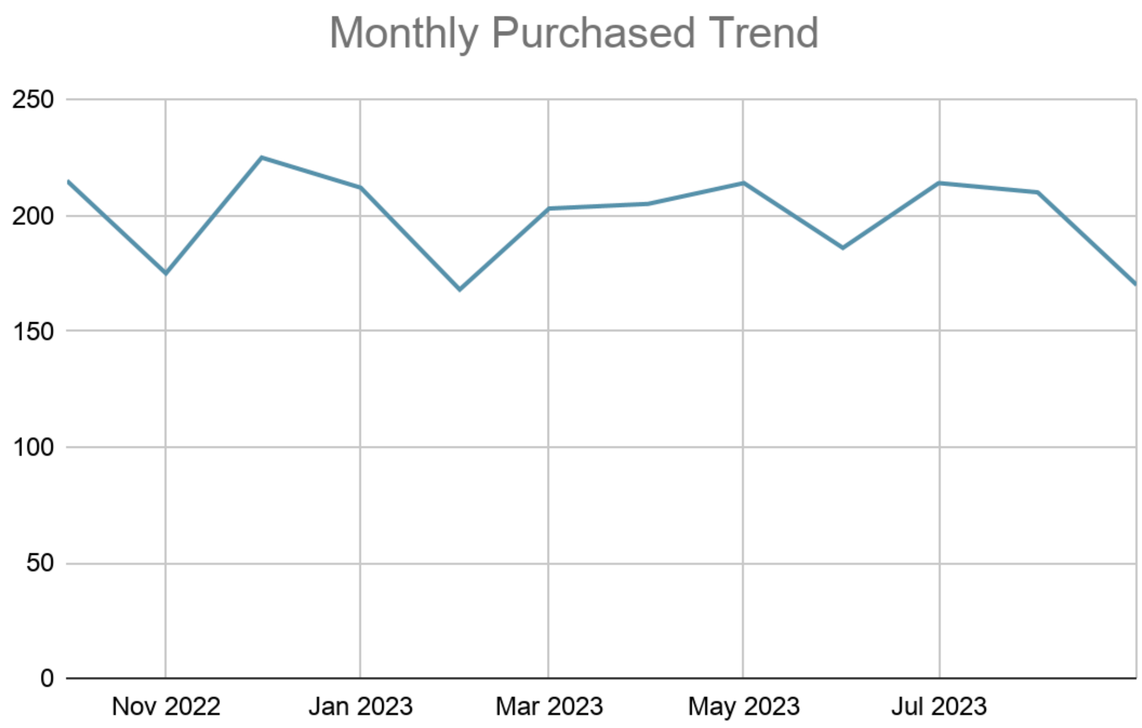
**Figure 2. Taiwan Market Potential Score by Category**




**Interpretation**

Figure 2 shows Taiwan Market Potential Score for different categories based on purchase frequency, average rating, and average purchase amount. The Bags & Accessories category had the highest score overall, showing the best business potential compared to other categories. The Footwear category had fewer purchases compared to Tops, but the high average purchase amount helped to make it have a high score. It shows that using several performance metrics can give a fairer evaluation than just using purchase frequency.

**Figure 3. Monthly Purchase Trend**

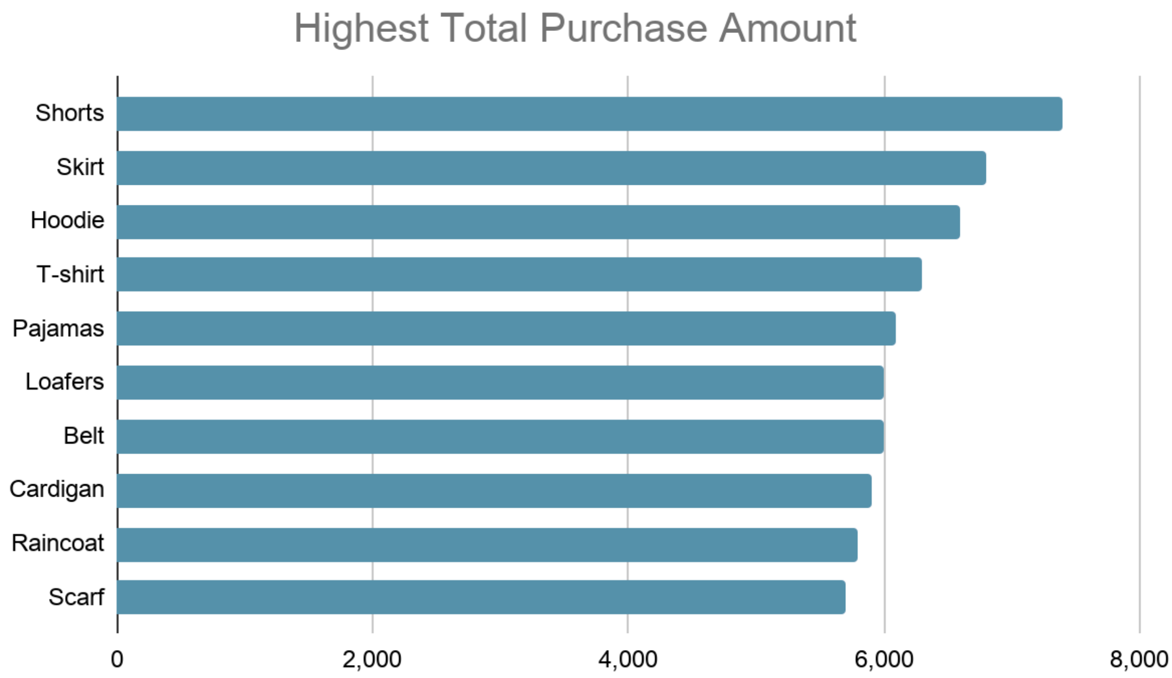



**Note: October 2023 represents an incomplete month because the dataset only contains transactions through October 1, 2023.**

**Interpretation**

Figure 3 provides data on monthly purchase behavior during the research period. In general, purchase volume has been fairly steady, with minor variations throughout the year. The lowest levels of purchase have occurred in November 2022, February 2023, and September 2023. The dramatic drop in October 2023 must be considered carefully since it reflects only an incomplete month, not a drop in consumer demand.

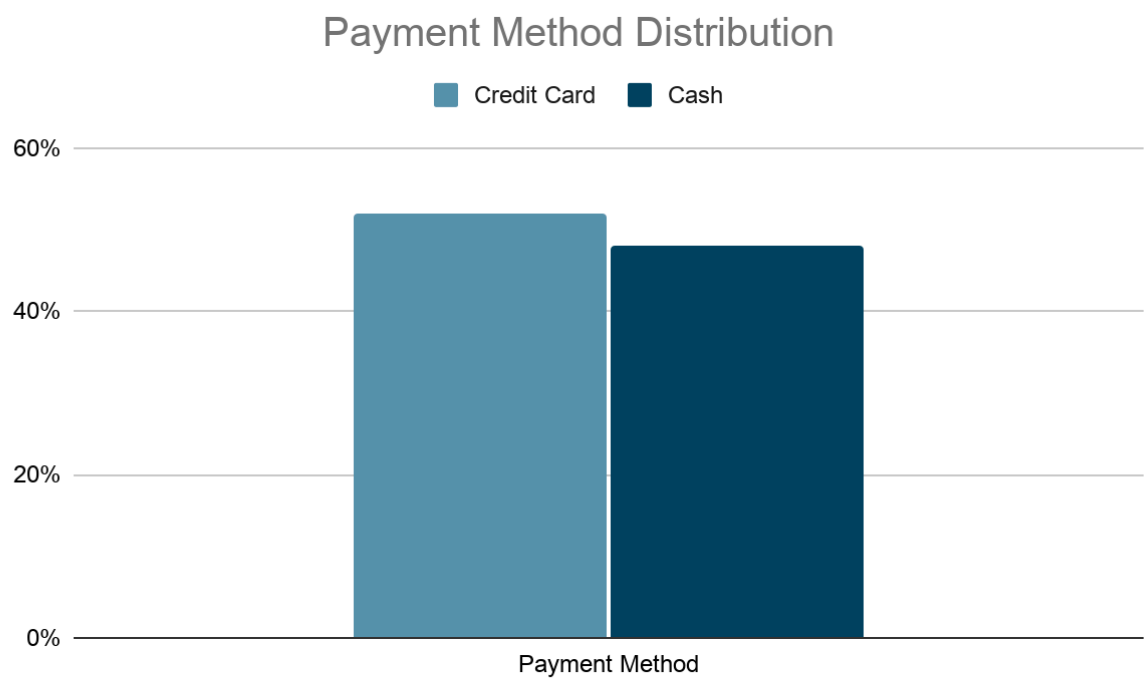
**Figure 4. Highest Total Purchase Amount**




**Interpretation**

Figure 4 shows the ten products that have the highest purchase totals individually. The product types Shorts, Skirts, and Hoodies had the largest sales values out of all products. Although Shorts, Skirts, and Hoodies had the largest purchase totals, it cannot be assumed that purchase total is the only way to measure market demand. Other considerations include how often customers purchase and their satisfaction levels.

**Figure 5. Payment Method Distribution**




**Interpretation**

In Figure 5, the payment methods adopted by customers are compared. It is clear from the results obtained that the use of credit cards was marginally higher than the use of cash by customers. This means that consumers feel at ease when using both types of payments. From the perspective of cross-border e-commerce firms, the adoption of secure online payments is crucial.

# 9. Taiwan Market Potential Score

The Taiwan Market Potential Score is a basic weighted index created to compare product types in terms of their business potential. This index is not designed to forecast any future sales in Taiwan. The Taiwan Market Potential Score is meant to be a decision-making aid that integrates several measures into one number.

The score is calculated using the following formula:

**Score = 45% Purchase Frequency + 30% Average Rating + 25% Average Purchase Amount**

The weighting method balances market demands, customer satisfaction, and purchase values to serve as a realistic model in evaluating different product categories.

- **Purchase Frequency (45%)** is assigned the highest weighting as it is the one that shows consumer demands and buying interests. The more frequently the category is bought, the higher the likelihood of market acceptance.

- **Average Rating (30%)** is assigned the second highest weighting because customer satisfaction determines the quality of the product and its future market success.

- **Average Purchase Amount (25%)** gets the least weighting. Even though more money may mean higher profits, not all expensive products are ideal for border crossing business.

**Figure 2 presents the Taiwan Market Potential Score for each product category.**

From the results, it is apparent that the category which performed the best was Bags & Accessories since it has the most outstanding average score resulting from high purchase frequency and stability in terms of customer ratings and purchase value. Footwear comes second because of its relatively high average purchase value and customer ratings even though it records a lower number of purchases compared to Tops. Despite Tops having a high customer demand, the relatively low average rating makes its score lower compared to the other categories.

## 10. Key Findings

1. **Bags & Accessories is the strongest product category**

   This product segment has the highest purchase rate and the highest market potential score. This includes useful cross-border goods like handbags, wallets, backpacks, belts, hats, sunglasses, scarves, and socks.

2. **Footwear has strong potential despite lower purchase volume**

   Footwear is not at the top of purchase numbers, but it does well in terms of average ratings and purchases. This implies that footwear may be useful for selective testing purposes in border trade.

3. **Tops have broad demand but need careful product selection**

   Tops are very high when it comes to purchases. Nonetheless, tops have a relatively low average rating compared to the other categories, implying that quality, design, fit, and sizes may be quite important.

4. **Monthly sales are relatively stable, with lower purchase activity in February, November, September, and the incomplete October 2023 period**

   The month-by-month trend can help support basic inventory planning and seasonal selection decisions.

5. **Payment method distribution is balanced**

   Cash and credit card payments accounted for a similar number of transactions, with credit card payments being slightly more prevalent, suggesting that digital payment readiness remains an important factor in e-commerce design.

# 11. Business Recommendations for Taiwan Market

1. **Prioritize accessories as the first product testing category**

   Accessories are simpler to measure in size, ship, and sell online compared to a lot of fitted apparel. On account of the findings, Bags & Accessories should be considered priority categories for testing in Taiwan.

2. **Test footwear selectively**

   Selectively test footwear. Footwear is a high potential category but could have problems with sizing and risks with returns.

3. **Use tops as a volume category but manage size risk**

   Treat Tops as a volume category but be mindful of the sizing challenge.
   Tops are an interesting category that could appeal to a wide audience but fit and style differences could play a role in customer satisfaction.

4. **Add brand level data before making brand recommendations**

   Include brand equity level analysis before recommending brands.
   At this moment, transaction data in the dataset does not contain brand equity and countries of origin information as well as sub-sector information. It should be considered in the next iteration of the project.

5. **Build a dashboard for ongoing product selection**

   Develop an interactive Tableau dashboard for selecting products. Creating a dashboard in Tableau will enable to compare categories by demand, average rating, average purchase, and potential at once.

## 12. Limitations

- The first database lacks information on brands, country, gender, and consumer characteristics unique to Taiwan.

- Purchase frequency serves as a measure of demand since there is no information on quantities sold.

- The Taiwan Market Potential Score is a decision-making index, and not a sales prediction.

- There were some big purchase frequencies that were considered outliers because they did not fit within the main range of values.

- Conclusions on the brand level need an uploaded database of brands or other sources of brand information.

Furthermore, the database is based on past transactions and does not indicate any specific consumer preference in Taiwan. The results are more likely to be used as a decision-making tool and not as an actual representation of the demand in the Taiwanese market. In future studies, there needs to be the inclusion of consumer data, brand details, and market indicators specific to Taiwan.

## 13. Future Improvements

- Import an entire Global Fashion Brands CSV to sort out brands on the basis of equity and growth.

- Include the consumer survey or search trend information of Taiwan to support your findings.

- Improve Taiwan-specific content using Google Trends or marketplace search queries.

- Develop a Tableau dashboard with filter options including category, payment type, month, and ratings.

- Apply forecasting techniques once long-term sales time series data is acquired.

# 14. Conclusion

Through this research project, we have seen how transactional data collected in retail business can be analyzed and transformed into meaningful business insights through data analytics. Based on analysis of frequency of purchase, customer ratings and the amount of purchases, the product categories which are more likely to perform well in cross-border retail targeting the Taiwanese market were identified.

Currently, our dataset does not contain any information about brands or consumer behavior in the Taiwan market; however, the developed score system to evaluate Taiwan market potential for each product category will provide a practical framework for comparison of various product categories based on multiple indicators of their performance.

In general, our project has provided us with an analytical model that can be expanded through integration of global datasets on fashion brands and local consumer preference data.

## References


- Fashion Retail Sales Dataset (Kaggle)
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- Global Fashion Brands (Kaggle)
  https://www.kaggle.com/code/rajatraj0502/global-fashion-brands

- Global Fashion Brands Dataset description (Mendeley Data)
  https://www.kaggle.com/datasets/jocelyndumlao/global-fashion-brands
  https://data.mendeley.com/datasets/7fg8835b4g